Scalar, vector, and tensor spherical harmonics

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I. SPHERICAL HARMONICS AND THEIR PROPERTIES

We define the spherical harmonics $Y_{lm}(\Omega)$ as

$$Y_{lm}(\Omega) = (-1)^m \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\vartheta) e^{\imath m\varphi}; \tag{1}$$

the Condon-Shortley phase eases the use of raising and lowering operators. It is possible to absorb it in the definition of the associated Legendre polynomials $P_l^m(x)$. The spherical harmonics are orthonormal,

$$\int \mathrm{d}\Omega\, Y_{lm}(\Omega) Y^*_{pq}(\Omega) = \delta_{lp}\delta_{mq}. \tag{2}$$

Additionally, $Y^*_{lm} = (-1)^m Y_{l\underline{m}}$, where $\underline{m} \equiv -m$. Spherical harmonics posses parity of $\mathcal{P}Y_{lm} = (-1)^l Y_{lm}$, where the parity operator transform $\boldsymbol{r} \to -\boldsymbol{r}$ or, equivalently, $(\vartheta, \varphi) \to (\vartheta - \pi, \varphi + \pi)$. Under a rotation $\mathcal{R}$, transforming $\boldsymbol{r} \to \boldsymbol{r}'$, the harmonics transform as $\mathcal{R}Y_{lm} = Y_{lm}(\boldsymbol{r}') = \sum_{m'} D^{(l)*}_{mm'}(\mathcal{R}) Y_{lm'}(\boldsymbol{r})$.

A. Useful properties and relations

In evaluating the vector spherical harmonics numerically, we need to pay attention to the terms with $1/\sin\vartheta$ and $\partial_\vartheta Y_{lm}$, as these are unstable close to the poles [1]. A useful relation for us will be

$$\begin{aligned}
\cot\vartheta Y_{lm} &= c(l,m)\, e^{\imath m\varphi} \frac{x}{\sqrt{1-x^2}} P_l^m(x) \\
&= c(l,m)\, e^{\imath m\varphi} \left(\frac{-1}{2m}\right) \left[P_l^{m+1}(x) + (l-m+1)(l+m)P_l^{m-1}\right] \\
&= \left(\frac{-1}{2m}\right) \left[e^{-\imath\varphi}\frac{c(l,m)}{c(l,m+1)} Y_{lm+1} + (l-m+1)(l+m)e^{\imath\varphi}\frac{c(l,m)}{c(l,m-1)} Y_{lm-1}\right] \\
&= -\frac{1}{2m}\left[e^{-\imath\varphi}\sqrt{(l-m)(l+m+1)} Y_{lm+1} + e^{\imath\varphi}\sqrt{(l-m+1)(l+m)} Y_{lm-1}\right].
\end{aligned} \tag{3}$$

This relation does not mix between l. Similarly, we can derive

$$\begin{aligned}
\frac{Y_{lm}}{\sin\vartheta} &= c(l,m)\, e^{\imath m\varphi} \frac{1}{\sqrt{1-x^2}} P_l^m(x) \\
&= c(l,m)\, e^{\imath m\varphi} \left(\frac{-1}{2m}\right) \left[P_{l-1}^{m+1}(x) + (l+m-1)(l+m)P_{l-1}^{m-1}\right] \\
&= \left(\frac{-1}{2m}\right) \left[e^{-\imath\varphi}\frac{c(l,m)}{c(l-1,m+1)} Y_{l-1m+1} + (l+m-1)(l+m)e^{\imath\varphi}\frac{c(l,m)}{c(l-1,m-1)} Y_{l-1m-1}\right] \\
&= -\frac{1}{2m}\sqrt{\frac{2l+1}{2l-1}}\left[e^{-\imath\varphi}\sqrt{(l-m)(l-m-1)} Y_{l-1m+1} + e^{\imath\varphi}\sqrt{(l+m)(l+m-1)} Y_{l-1m-1}\right],
\end{aligned} \tag{4}$$

which lowers l. This expression can be written in an alternative way, raising instead of lowering l:

$$\begin{aligned}
\frac{Y_{lm}}{\sin\vartheta} &= c(l,m) e^{\imath m\varphi} \frac{P_l^m(x)}{\sqrt{1-x^2}} \\
&= -\frac{c(l,m)e^{\imath m\varphi}}{2m}\left[P_{l+1}^{m+1}(x) + (l-m+1)(l-m+2)P_{l+1}^{m-1}(x)\right] \\
&= -\frac{1}{2m}\left[e^{-\imath\varphi}\frac{c(l,m)}{c(l+1,m+1)} Y_{l+1,m+1} + (l-m+1)(l-m+2)e^{\imath\varphi}\frac{c(l,m)}{c(l+1,m-1)} Y_{l+1,m-1}\right] \\
&= -\frac{1}{2m}\sqrt{\frac{2l+1}{2l+3}}\left[e^{-\imath\varphi}\sqrt{(l+m+1)(l+m+2)} Y_{l+1m+1} + e^{\imath\varphi}\sqrt{(l-m+1)(l-m+2)} Y_{l+1m-1}\right].
\end{aligned} \tag{5}$$

The relation which uses lowered l might be more useful for computational purposes, as it lends itself to using an array of spherical harmonics only up to and including l.

For completeness sake, we notice that one may evaluate an expression with $1/\sin^2\vartheta$ as a combination of raising and lowering operations, thus obtaining an expression which does not change l:

$$\begin{aligned}\frac{Y_{lm}}{\sin^2\vartheta} &= \frac{1}{4m(m+1)}e^{-2\imath\varphi}\sqrt{(l-m)(l-m-1)(l+m+1)(l+m+2)}Y_{lm+2}\\ &+\frac{1}{4m(m-1)}e^{2\imath\varphi}\sqrt{(l+m)(l+m-1)(l-m+1)(l-m+2)}Y_{lm-2}\\ &+\frac{1}{2}\frac{l(l+1)+m^2-1}{m^2-1}Y_{lm}.\end{aligned} \tag{6}$$

For calculations, it is good to know also the derivatives of spherical harmonics. For the derivative with respect to φ, we have

$$\partial_\varphi Y_{lm} = \imath m Y_{lm}, \tag{7}$$

and thus also $\partial_{\varphi\varphi}Y_{lm} = -m^2 Y_{lm}$. The derivative with respect to ϑ is more involved (via Wolfram),

$$\begin{aligned}\partial_\vartheta Y_{lm} &= m\cot\vartheta Y_{lm} + \sqrt{(l-m)(l+m+1)}\,e^{-\imath\varphi}Y_{lm+1}\\ &= \frac{1}{2}\left[e^{-\imath\varphi}\sqrt{(l-m)(l+m+1)}Y_{lm+1} - e^{\imath\varphi}\sqrt{(l-m+1)(l+m)}Y_{lm-1}\right],\end{aligned} \tag{8}$$

using previously derived expressions. Applying the same derivative once more, we get

$$\begin{aligned}\partial_{\vartheta\vartheta}Y_{lm} &= \frac{1}{4}e^{-2\imath\varphi}\sqrt{(l-m)(l+m+1)(l-m-1)(l+m+2)}Y_{lm+2}\\ &+\frac{1}{4}e^{2\imath\varphi}\sqrt{(l-m+1)(l+m)(l-m+2)(l+m-1)}Y_{lm-2}\\ &-\frac{1}{2}\left[l(l+1)-m^2\right]Y_{lm}.\end{aligned} \tag{9}$$

Hopefully, all the factors are correct; at least it can be easily double-checked. The mixed derivatives are easier; we have

$$\partial_{\vartheta\varphi}Y_{lm} = \frac{\imath m}{2}\left[e^{-\imath\varphi}\sqrt{(l-m)(l+m+1)}Y_{lm+1} - e^{\imath\varphi}\sqrt{(l-m+1)(l+m)}Y_{lm-1}\right] \tag{10}$$

with $\partial_{\varphi\vartheta}Y_{lm} = \partial_{\vartheta\varphi}Y_{lm}$.

B. Sanity check #1

The angular part of the Laplace operator in spherical coordinates is

$$\nabla^2 = \frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\partial_\vartheta) + \frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi} = \partial_{\vartheta\vartheta} + \cot\vartheta\partial_\vartheta + \frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}, \tag{11}$$

omitting the radial parts. We thus have

$$\begin{aligned}
\nabla^2 Y_{lm} = & \frac{1}{4} e^{-2\imath\varphi} \sqrt{(l-m)(l+m+1)(l-m-1)(l+m+2)} Y_{lm+2} \\
& + \frac{1}{4} e^{2\imath\varphi} \sqrt{(l-m+1)(l+m)(l-m+2)(l+m-1)} Y_{lm-2} \\
& - \frac{1}{2} \left[l(l+1) - m^2 \right] Y_{lm} \\
& - \frac{m}{4(m+1)} e^{-2\imath\varphi} \sqrt{(l-m)(l-m-1)(l+m+1)(l+m+2)} Y_{lm+2} \\
& - \frac{m}{4(m-1)} e^{2\imath\varphi} \sqrt{(l+m)(l+m-1)(l-m+1)(l-m+2)} Y_{lm-2} \\
& - \frac{m^2}{2} \frac{l(l+1) + m^2 - 1}{m^2 - 1} Y_{lm} \\
& - \frac{1}{4(m+1)} e^{-2\imath\varphi} \sqrt{(l-m)(l+m+1)(l-m-1)(l+m+2)} Y_{lm+2} \\
& + \frac{1}{4(m-1)} e^{2\imath\varphi} \sqrt{(l-m+1)(l+m)(l-m+2)(l+m-1)} Y_{lm-2} \\
& + \frac{1}{2} \frac{l(l+1)}{m^2 - 1} Y_{lm}. \qquad (12)
\end{aligned}$$

The terms with $m+2$ and $m-2$ spherical harmonics sum to zero. The rest sums up, much to our delight, to

$$\nabla^2 Y_{lm} = -l(l+1) Y_{lm}. \qquad (13)$$

C. Scalar multipole moments

Given a scalar function $\rho(\Omega)$ on a unit sphere, one can expand it in terms of scalar multipole moments as

$$\rho(\Omega) = \sum_{l,m} \rho_{lm} Y_{lm}(\Omega); \qquad (14)$$

the multipole expansion coefficients ρ_{lm} are obtained as

$$\rho_{lm} = \int \mathrm{d}\Omega \, \rho(\Omega) Y^*_{lm}(\Omega), \qquad (15)$$

due to the orthonormality of spherical harmonics. If the scalar function is composed of delta functions, $\rho(\Omega) = \sum_{k=1}^{N} \rho_k \delta(\Omega - \Omega_k)$, the multipole coefficients reduce to $\rho_{lm} = \sum_k \rho_k Y^*_{lm}(\Omega_k)$. One can proceed to define multipole magnitudes, purely real quantities, as

$$Q_l = \sqrt{\frac{4\pi}{2l+1} \sum_m |\rho_{lm}|^2}; \qquad (16)$$

the square of this is related to the spherical structure factor, $S_N(\ell) = Q_l^2/N$, as we know [2]. In this definition, the zero-th multipole moment corresponds to the total number of particles if ρ is the density, etc.

II. VECTOR SPHERICAL HARMONICS

We define three types of vector spherical harmonics (VSH) as

$$\boldsymbol{Y}_{lm} = \widehat{\boldsymbol{r}} \, Y_{lm}, \qquad (17)$$

$$\boldsymbol{\Psi}_{lm} = \frac{r}{\sqrt{l(l+1)}} \boldsymbol{\nabla} Y_{lm}, \qquad (18)$$

$$\boldsymbol{\Phi}_{lm} = \frac{1}{\sqrt{l(l+1)}} \boldsymbol{L} Y_{lm} = \frac{r}{\sqrt{l(l+1)}} \widehat{\boldsymbol{r}} \times \boldsymbol{\nabla} Y_{lm}, \qquad (19)$$

where $\widehat{\boldsymbol{r}}$ is the unit radial vector and $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{\nabla}$ is the angular momentum operator. This normalization differs from the commonly used one [3, 4] and is used to ensure orthonormality [5]:

$$\int \mathrm{d}\Omega\, \boldsymbol{Y}_{lm} \cdot \boldsymbol{Y}^*_{pq} = \delta_{lp}\delta_{mq}, \tag{20}$$

$$\int \mathrm{d}\Omega\, \boldsymbol{\Psi}_{lm} \cdot \boldsymbol{\Psi}^*_{pq} = \delta_{lp}\delta_{mq}, \tag{21}$$

$$\int \mathrm{d}\Omega\, \boldsymbol{\Phi}_{lm} \cdot \boldsymbol{\Phi}^*_{pq} = \delta_{lp}\delta_{mq}, \tag{22}$$

$$\int \mathrm{d}\Omega\, \boldsymbol{\Psi}_{lm} \cdot \boldsymbol{\Phi}^*_{pq} = 0, \tag{23}$$

$$\tag{24}$$

and so on. The first two VSH have electric-type parity, $\mathcal{P}\boldsymbol{Y}_{lm} = \mathcal{P}\boldsymbol{\Psi}_{lm} = (-1)^l$, while the third one has magnetic-type parity, $\mathcal{P}\boldsymbol{\Phi}_{lm} = (-1)^{l+1}$. Additionally, it holds that $\boldsymbol{\Psi}_{lm} = -\widehat{\boldsymbol{r}} \times \boldsymbol{\Phi}_{lm}$ and $\boldsymbol{\Phi}_{lm} = \widehat{\boldsymbol{r}} \times \boldsymbol{\Psi}_{lm}$.

(Maybe we can also write the VSH as $\boldsymbol{V}^{(\kappa)}$, where $\kappa = r, E, B$?)

A. Vector multipole moments

We start with a representation of N point dipoles $\boldsymbol{\mu}_k$ located on the surface of a unit sphere ($r = 1$):

$$\boldsymbol{\mu}(\Omega) = \sum_{k=1}^{N} \boldsymbol{\mu}_k \delta(\Omega - \Omega_k), \tag{25}$$

where $\Omega_k = (\vartheta_k, \varphi_k)$ are the positions of the dipoles. We can expand this density in terms of vector spherical harmonics as

$$\boldsymbol{\mu} = \sum_{l,m} \left[\mu^r_{lm}\boldsymbol{Y}_{lm} + \mu^E_{lm}\boldsymbol{\Psi}_{lm} + \mu^B_{lm}\boldsymbol{\Phi}_{lm}\right]. \tag{26}$$

The first type of spherical harmonics, $\boldsymbol{Y}_{lm}$, gives the radial component of the vector field on the sphere, whereas the other two types, $\boldsymbol{\Psi}_{lm}$ and $\boldsymbol{\Phi}_{lm}$, are the spheroidal and toroidal functions, respectively, and describe the in-plane (surface) dependence of the vector field [1, 3, 4]. The multipole expansion coefficients of the vector field $\boldsymbol{\mu}$ can be obtained as

$$\mu^r_{lm} = \sum_k \boldsymbol{\mu}_k \cdot \boldsymbol{Y}^*_{lm}(\Omega_k), \tag{27}$$

$$\mu^E_{lm} = \sum_k \boldsymbol{\mu}_k \cdot \boldsymbol{\Psi}^*_{lm}(\Omega_k), \tag{28}$$

$$\mu^B_{lm} = \sum_k \boldsymbol{\mu}_k \cdot \boldsymbol{\Phi}^*_{lm}(\Omega_k), \tag{29}$$

and equivalently for a continuous $\boldsymbol{\mu}$. One can imagine the definition of a multipole magnitude as

$$Q^{(\kappa)}_l = \sqrt{\frac{4\pi}{2l+1}\sum_m |\mu^{(\kappa)}_{lm}|^2}, \tag{30}$$

where $\kappa = r, E, B$.

B. Implementation notes

For the gradient operator in spherical coordinates and the angular momentum $\boldsymbol{L} = r\widehat{\boldsymbol{r}} \times \boldsymbol{\nabla}$, we have

$$r\boldsymbol{\nabla} = r\widehat{\boldsymbol{r}}\,\partial_r + \widehat{\boldsymbol{\vartheta}}\,\partial_\vartheta + \widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_\varphi, \tag{31}$$

$$\boldsymbol{L} = \widehat{\boldsymbol{\varphi}}\,\partial_\vartheta - \widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\theta}\partial_\varphi. \tag{32}$$

Thus,

$$\boldsymbol{\Psi}_{lm} = \frac{1}{\sqrt{l(l+1)}} \left(\widehat{\boldsymbol{\vartheta}}\, \partial_\vartheta Y_{lm} + \widehat{\boldsymbol{\varphi}}\, \frac{\partial_\varphi Y_{lm}}{\sin\vartheta} \right), \tag{33}$$

$$\boldsymbol{\Phi}_{lm} = \frac{1}{\sqrt{l(l+1)}} \left(-\widehat{\boldsymbol{\vartheta}}\, \frac{\partial_\varphi Y_{lm}}{\sin\vartheta} + \widehat{\boldsymbol{\varphi}}\, \partial_\vartheta Y_{lm} \right). \tag{34}$$

To evaluate the VSH, we need to know only the expressions for $\partial_\vartheta Y_{lm}$, given in Eq. (8), and for $\partial_\varphi Y_{lm}/\sin\vartheta$, which can be obtained by combining Eqs. (7) and (4). Thus, if $\boldsymbol{\mu}_k = (\mu_r^{(k)}, \mu_\vartheta^{(k)}, \mu_\varphi^{(k)})$,

$$\mu_{lm}^r = \sum_k \mu_r^{(k)} Y_{lm}^*(\Omega_k), \tag{35}$$

$$\mu_{lm}^E = \frac{1}{\sqrt{l(l+1)}} \sum_k \left[\mu_\vartheta^{(k)} \partial_\vartheta Y_{lm}^*(\Omega_k) + \mu_\varphi^{(k)} \frac{\partial_\varphi Y_{lm}^*(\Omega_k)}{\sin\vartheta} \right], \tag{36}$$

$$\mu_{lm}^B = \frac{1}{\sqrt{l(l+1)}} \sum_k \left[-\mu_\vartheta^{(k)} \frac{\partial_\varphi Y_{lm}^*(\Omega_k)}{\sin\vartheta} + \mu_\varphi^{(k)} \partial_\vartheta Y_{lm}^*(\Omega_k) \right]. \tag{37}$$

III. TENSOR SPHERICAL HARMONICS

This section mostly follows Zerilli [6]. This is a summary of the paper and the equations were not checked for $\pm$ accuracy, which is a TODO. Note that Thorne [5] points out a couple of errors in the Zerilli paper, but these all seem to be corrected in the version currently published online.

Given two tensors $\underline{\underline{\mathrm{T}}} = T^{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j$ and $\underline{\underline{\mathrm{S}}} = S^{kl} \boldsymbol{e}_k \otimes \boldsymbol{e}_l$, where $\boldsymbol{e}_i$ constitute an orthonormal basis, we define the dot product of the two tensors as

$$\underline{\underline{\mathrm{T}}} : \underline{\underline{\mathrm{S}}} = T^{ij} S_{ij}. \tag{38}$$

If Y_{LM} is a spherical harmonic, we say that $Y_{LM} \boldsymbol{e}_i \boldsymbol{e}_j$ is orthonormal in the inner product

$$(\underline{\underline{\mathrm{T}}}, \underline{\underline{\mathrm{S}}}) = \int \mathrm{d}\Omega\, \underline{\underline{\mathrm{T}}}^* : \underline{\underline{\mathrm{S}}}. \tag{39}$$

Restricting our discussion to symmetric tensors with trace, it can be shown that there are six of them. We can obtain them by operating on Y_{LM} with $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{\nabla}$, $\boldsymbol{\nabla}$, and $\boldsymbol{e}_r$. Thus, we obtain expressions for the symmetric parts of $\boldsymbol{\nabla}\boldsymbol{\nabla} Y_{LM}$, $\boldsymbol{L}\boldsymbol{L} Y_{LM}$, $\boldsymbol{e}_r \boldsymbol{e}_r Y_{LM}$, $\boldsymbol{e}_r \boldsymbol{\nabla} Y_{LM}$, $\boldsymbol{L}\boldsymbol{\nabla} Y_{LM}$, and $\boldsymbol{e}_r \boldsymbol{L} Y_{LM}$. The first four tensors have parity $(-1)^L$ ("magnetic") and the last two have parity $(-1)^{L+1}$ ("electric"), and together they suffice to represent an arbitrary symmetric tensor on the sphere.

The paper goes on to derive explicit expressions of these tensors in spherical coordinates [Eqs. (7a-f)]. The six tensors can be then further separated into three tensors orthogonal to the sphere and three tensors tangent to the sphere. The three tensors orthogonal to the sphere (one electric and two magnetic) are

$$\underline{\underline{\mathrm{a}}}_{LM} = [\boldsymbol{e}_r \boldsymbol{e}_r Y_{LM}]_s, \tag{40}$$

$$\underline{\underline{\mathrm{b}}}_{LM} = \sqrt{\frac{2}{L(L+1)}}\, r\, [\boldsymbol{e}_r \boldsymbol{\nabla} Y_{LM}]_s, \tag{41}$$

$$\underline{\underline{\mathrm{c}}}_{LM} = \sqrt{\frac{2}{L(L+1)}} [\boldsymbol{e}_r \boldsymbol{L} Y_{LM}]_s. \tag{42}$$

The three tensors tangent to the sphere (one electric and two magnetic) are

$$\underline{\underline{\mathrm{d}}}_{LM} = \sqrt{\frac{2}{L(L+1)(L-1)(L+2)}}\, r \left\{ [\boldsymbol{L}\boldsymbol{\nabla} Y_{LM}]_s + \frac{1}{r} [\boldsymbol{e}_r \boldsymbol{L} Y_{LM}]_s \right\}, \tag{43}$$

$$\underline{\underline{\mathrm{f}}}_{LM} = \sqrt{\frac{1}{2L(L+1)(L-1)(L+2)}} (\underline{\underline{\mathrm{e}}}_{LM} + \underline{\underline{\mathrm{h}}}_{LM}), \tag{44}$$

$$\underline{\underline{\mathrm{g}}}_{LM} = -\frac{1}{\sqrt{2} L(L+1)} (\underline{\underline{\mathrm{e}}}_{LM} - \underline{\underline{\mathrm{h}}}_{LM}), \tag{45}$$

where

$$\underline{\underline{\mathrm{e}}}_{lm} = r^2 \left\{ [\boldsymbol{\nabla}\boldsymbol{\nabla} Y_{LM}]_s + \frac{2}{r}[\boldsymbol{e}_r \boldsymbol{\nabla} Y_{LM}]_s \right\}, \tag{46}$$

$$\underline{\underline{\mathrm{h}}}_{lm} = [\boldsymbol{L}\boldsymbol{L} Y_{LM}]_s + r\,[\boldsymbol{e}_r \boldsymbol{\nabla} Y_{LM}]_s. \tag{47}$$

Thus, we can write any symmetric tensor $\underline{\underline{\mathrm{T}}}$ on the sphere as

$$\underline{\underline{\mathrm{T}}} = \sum_{L,M} \left(A_{LM}\underline{\underline{\mathrm{a}}}_{LM} + B_{LM}\underline{\underline{\mathrm{b}}}_{LM} + C_{LM}\underline{\underline{\mathrm{c}}}_{LM} + D_{LM}\underline{\underline{\mathrm{d}}}_{LM} + F_{LM}\underline{\underline{\mathrm{f}}}_{LM} + G_{LM}\underline{\underline{\mathrm{g}}}_{LM} \right), \tag{48}$$

where, I imagine, we have

$$A_{LM} = (\underline{\underline{\mathrm{T}}}, \underline{\underline{\mathrm{a}}}_{LM}) = \int \mathrm{d}\Omega\, \underline{\underline{\mathrm{T}}}^* : \underline{\underline{\mathrm{a}}}_{LM}. \tag{49}$$

Since we know the six tensors in spherical coordinates, we can extract the multipole coefficients from an arbitrary symmetric tensor in spherical coordinates.

Assume now we have a distribution of N quadrupoles on a sphere,

$$Q_{ij}(\Omega) = \sum_{k=1}^{N} Q_{ij}^{(k)} \delta(\Omega - \Omega_k). \tag{50}$$

We can then expand it in terms of 6 basis functions defined above (which we now define simply as $t_{ij}^{(\mu)}$, $\mu = 1, \ldots, 6$:

$$Q_{ij} = \sum_{l,m} \sum_{\mu} \gamma_{lm}^{(\mu)} t_{ij,lm}^{(\mu)} \tag{51}$$

For each μ, we thus have due to orthogonality of $t_{ij,lm}^{(\mu)}$:

$$\gamma_{lm}^{(\mu)} = \int \mathrm{d}\Omega\, Q_{ij}^*(\Omega) t_{ij,lm}^{(\mu)}(\Omega) = \sum_{k=1}^{N} Q_{ij}^{(k)*} t_{ij,lm}^{(\mu)}(\Omega_k) \tag{52}$$

We now list all six tensors explicitly in spherical coordinates (dependence on Ω is considered implicitly):

$$t_{lm}^{(0)} = \begin{pmatrix} Y_{lm} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{53}$$

Thus, we have trivially

$$\gamma_{lm}^{(0)} = \sum_k Q_{rr}^{(k)*} Y_{lm}(\Omega_k) \tag{54}$$

For the second tensor, we have

$$t_{lm}^{(1)} = \frac{r}{\sqrt{2l(l+1)}} \begin{pmatrix} 0 & \partial_\vartheta Y_{lm} & \partial_\varphi Y_{lm} \\ * & 0 & 0 \\ * & * & 0 \end{pmatrix} \tag{55}$$

Here, asterisks denote symmetric parts of the tensor. The derivatives are known from before [Eqs. (**??**) and (**??**)] and we can again express them solely in terms of spherical harmonics, evaluated at Ω_k. One can thus write

$$\gamma_{lm}^{(1)} = \frac{r}{\sqrt{2l(l+1)}} \sum_k \left[Q_{r\vartheta}^{(k)*} \partial_\vartheta Y_{lm}(\Omega)\Big|_{\Omega_k} + Q_{r\varphi}^{(k)*} \partial_\varphi Y_{lm}(\Omega)\Big|_{\Omega_k} \right] \tag{56}$$

The third tensor is similar, and reads

$$t_{lm}^{(2)} = \frac{\imath r}{\sqrt{2l(l+1)}} \begin{pmatrix} 0 & \frac{1}{\sin\vartheta}\partial_\varphi Y_{lm} & -\sin\vartheta \partial_\vartheta Y_{lm} \\ * & 0 & 0 \\ * & * & 0 \end{pmatrix} \tag{57}$$

We again know how to express the term $(1/\sin\vartheta)\partial_\varphi Y_{lm}$ solely in terms of spherical harmonics [Eq. (**??**)], which allows us to write

$$\gamma^{(2)}_{lm} = \frac{\imath r}{\sqrt{2l(l+1)}} \sum_k \left[Q^{(k)*}_{r\vartheta} \frac{1}{\sin\vartheta} \partial_\varphi Y_{lm}(\Omega)\Big|_{\Omega_k} - Q^{(k)*}_{r\varphi} \sin\vartheta \partial_\varphi Y_{lm}(\Omega)\Big|_{\Omega_k} \right] \tag{58}$$

The next three expressions are somewhat more involved. The fourth tensor, obtained after summation of two matrices, is thus

$$t^{(3)}_{lm} = -\frac{\imath r^2}{\sqrt{2l(l+1)(l-1)(l+2)}} \begin{pmatrix} 0 & 0 & 0 \\ * & -\frac{1}{\sin\vartheta} X_{lm} & \sin\vartheta W_{lm} \\ * & * & \sin\vartheta X_{lm} \end{pmatrix} \tag{59}$$

Here, we have

$$X_{lm} = 2\partial_\varphi \left(\partial_\vartheta - \cot\vartheta\right) Y_{lm}, \tag{60}$$

$$W_{lm} = \left(\partial_{\vartheta\vartheta} - \cot\vartheta\partial_\vartheta - \frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}\right) Y_{lm}. \tag{61}$$

For the first expression, we can obtain

$$X_{lm} = \imath \left[(m+2)\sqrt{(l-m)(l+m+1)} e^{-\imath\varphi} Y_{lm+1} - (m-2)\sqrt{(l+m)(l-m+1)} e^{\imath\varphi} Y_{lm-1} + 2m\cos\vartheta \frac{Y_{lm}}{\sin\vartheta} \right]. \tag{62}$$

We have opted to expand the $\cot\vartheta$, as $Y_{lm}/\sin\vartheta$ can again be expressed solely in terms of spherical harmonics [Eq. (**??**)]. For the second expression, we only note its parts:

$$\partial_{\varphi\varphi} Y_{lm} = -m^2 Y_{lm}, \tag{63}$$

and the $\cot\vartheta Y_{lm}$ term can again be expressed solely in terms of spherical harmonics via Eq. (**??**). The double derivative is more involved, but it can be shown to be

$$\partial_{\vartheta\vartheta} Y_{lm} = -\frac{1}{2}\left[l(l+1) - m^2\right] Y_{lm} + \tag{64}$$

$$+ \frac{1}{4}\sqrt{(l-m)(l+m+1)(l-m+1)(l+m+2)} e^{-2\imath\varphi} Y_{lm+2} + \tag{65}$$

$$+ \frac{1}{4}\sqrt{(l+m)(l-m+1)(l+m-1)(l-m+2)} e^{2\imath\varphi} Y_{lm-2}.$$

For the last two tensors, we need to first evaluate some other tensors. To start,

$$\underline{\underline{\mathrm{e}}}_{lm} = \begin{pmatrix} 0 & 0 & 0 \\ * & r^2\partial_{\vartheta\vartheta} Y_{lm} & \frac{1}{2} r^2 X_{lm} \\ * & * & -r^2\sin^2\vartheta\left[\partial_{\vartheta\vartheta} Y_{lm} + l(l+1) Y_{lm}\right] \end{pmatrix} \tag{66}$$

and

$$\underline{\underline{\mathrm{h}}}_{lm} = \begin{pmatrix} 0 & 0 & 0 \\ * & r^2\left[\partial_{\vartheta\vartheta} Y_{lm} + l(l+1) Y_{lm}\right] & \frac{1}{2} r^2 X_{lm} \\ * & * & -r^2\sin^2\vartheta\partial_{\vartheta\vartheta} Y_{lm} \end{pmatrix} \tag{67}$$

From here, we can determine

$$\underline{\underline{\mathrm{e}}}_{lm} + \underline{\underline{\mathrm{h}}}_{lm} = \begin{pmatrix} 0 & 0 & 0 \\ * & r^2\left[2\partial_{\vartheta\vartheta} Y_{lm} + l(l+1) Y_{lm}\right] & r^2 X_{lm} \\ * & * & -r^2\sin^2\vartheta\left[2\partial_{\vartheta\vartheta} Y_{lm} + l(l+1) Y_{lm}\right] \end{pmatrix} \tag{68}$$

and

$$\underline{\underline{\mathrm{e}}}_{lm} - \underline{\underline{\mathrm{h}}}_{lm} = \begin{pmatrix} 0 & 0 & 0 \\ * & -r^2\, l(l+1) Y_{lm} & 0 \\ * & * & -r^2\sin^2\vartheta\, l(l+1) Y_{lm} \end{pmatrix} \tag{69}$$

We thus have

$$t^{(4)}_{lm} = \frac{r^2}{\sqrt{2l(l+1)(l-1)(l+2)}} \begin{pmatrix} 0 & 0 & 0 \\ * & 2\partial_{\vartheta\vartheta} Y_{lm} + l(l+1) Y_{lm} & X_{lm} \\ * & * & -\sin^2\vartheta \left[2\partial_{\vartheta\vartheta} Y_{lm} + l(l+1) Y_{lm}\right] \end{pmatrix} \tag{70}$$

and

$$t^{(5)}_{lm} = \frac{r^2}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 0 \\ * & Y_{lm} & 0 \\ * & * & \sin^2\vartheta\, Y_{lm} \end{pmatrix} \tag{71}$$

For the last case, we see that we clearly have

$$\gamma^{(5)}_{lm} = \frac{r^2}{\sqrt{2}} \sum_k \left[Q^{(k)*}_{\vartheta\vartheta} Y_{lm}(\Omega_k) + Q^{(k)*}_{\varphi\varphi} \sin^2\vartheta Y_{lm}(\Omega_k) \right] \tag{72}$$

A. Independent derivation to find the source of discrepancies

Recap what we will need:

$$r\boldsymbol{\nabla} = r\widehat{\boldsymbol{r}}\partial_r + \widehat{\boldsymbol{\vartheta}}\partial_\vartheta + \widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_\varphi, \tag{73}$$

$$\boldsymbol{L} = \widehat{\boldsymbol{\varphi}}\partial_\vartheta - \widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\vartheta}\partial_\varphi, \tag{74}$$

$$\partial_\vartheta \widehat{\boldsymbol{r}} = \widehat{\boldsymbol{\vartheta}}, \tag{75}$$

$$\partial_\varphi \widehat{\boldsymbol{r}} = \sin\vartheta \widehat{\boldsymbol{\varphi}}, \tag{76}$$

$$\partial_\vartheta \widehat{\boldsymbol{\vartheta}} = -\widehat{\boldsymbol{r}}, \tag{77}$$

$$\partial_\vartheta \widehat{\boldsymbol{\varphi}} = 0, \tag{78}$$

$$\partial_\varphi \widehat{\boldsymbol{\vartheta}} = \cos\vartheta \widehat{\boldsymbol{\varphi}}, \tag{79}$$

$$\partial_\varphi \widehat{\boldsymbol{\varphi}} = -\sin\vartheta \widehat{\boldsymbol{r}} - \cos\vartheta \widehat{\boldsymbol{\vartheta}}. \tag{80}$$

$\underline{\underline{\mathrm{a}}}$ is straight forward as it has no derivatives. I'll ignore normalization constants for now. For the next tensor, we already see a discrepancy, depending if we symmetrize the operator or the resulting matrix (which feels like cheating):

$$\underline{\underline{\mathrm{b}}}' = \tfrac{1}{2}(\widehat{\boldsymbol{r}}(r\boldsymbol{\nabla}) + (r\boldsymbol{\nabla})\widehat{\boldsymbol{r}})Y = \tfrac{1}{2}(\widehat{\boldsymbol{r}}(r\boldsymbol{\nabla})Y + (r\boldsymbol{\nabla}Y)\widehat{\boldsymbol{r}} + (r\boldsymbol{\nabla}\widehat{\boldsymbol{r}})Y) \tag{81}$$

$$\underline{\underline{\mathrm{b}}}' = \tfrac{1}{2}(\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\vartheta}}\partial_\vartheta Y + \widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_\varphi Y + \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{r}}\partial_\vartheta Y + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{r}}\frac{1}{\sin\vartheta}\partial_\varphi Y + (\widehat{\boldsymbol{\vartheta}}\partial_\vartheta\widehat{\boldsymbol{r}} + \widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_\varphi\widehat{\boldsymbol{r}})Y \tag{82}$$

$$\underline{\underline{\mathrm{b}}}' = [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\vartheta}}]_s\partial_\vartheta Y + [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}]_s\frac{1}{\sin\vartheta}\partial_\varphi Y + \tfrac{1}{2}(\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}} + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}})Y \tag{83}$$

as opposed to

$$\underline{\underline{\mathrm{b}}} = [\widehat{\boldsymbol{r}}(r\boldsymbol{\nabla})Y]_s = [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\vartheta}}]_s\partial_\vartheta Y + [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}]_s\frac{1}{\sin\vartheta}\partial_\varphi Y \tag{84}$$

However, in both cases, I get an extra $\sin\vartheta$ in $[\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}]_s$ compared to Eq. 55.

Next is $\underline{\underline{\mathrm{c}}}$. Again, trying both ways.

$$\underline{\underline{\mathrm{c}}}' = \tfrac{1}{2}(\widehat{\boldsymbol{r}}\boldsymbol{L} + \boldsymbol{L}\widehat{\boldsymbol{r}})Y = \tfrac{1}{2}(\widehat{\boldsymbol{r}}\boldsymbol{L}Y + (\boldsymbol{L}Y)\widehat{\boldsymbol{r}} + (\boldsymbol{L}\widehat{\boldsymbol{r}})Y) \tag{85}$$

$$\underline{\underline{\mathrm{c}}}' = [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}]_s\partial_\vartheta Y - [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\vartheta}}]_s\frac{1}{\sin\vartheta}\partial_\varphi Y + \tfrac{1}{2}(\widehat{\boldsymbol{\varphi}}\partial_\vartheta\widehat{\boldsymbol{r}} - \frac{1}{\sin\vartheta}\widehat{\boldsymbol{\vartheta}}\partial_\varphi\widehat{\boldsymbol{r}})Y = [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}]_s\partial_\vartheta Y - [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\vartheta}}]_s\frac{1}{\sin\vartheta}\partial_\varphi Y + \tfrac{1}{2}(\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}} - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}})Y \tag{86}$$

Here, the anticommutator isn't symmetric, so symmetrizing after application of the operator seems reasonable. In that case, we get:

$$\underline{\underline{\mathrm{c}}} = [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}]_s \partial_\vartheta Y - [\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\vartheta}}]_s \frac{1}{\sin\vartheta}\partial_\varphi Y \tag{87}$$

This again has an one less $\sin\vartheta$ in the $\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}$ term compared to Eq. 57. Is our basis not orthonormalized correctly? The sign is inconsequential, Eq. 57 even has an imaginary unit there.

Now for the hard part. Consider this:

$$\begin{aligned}
[\boldsymbol{L}(r\nabla)Y]_s &= [\boldsymbol{L}(\widehat{\boldsymbol{\vartheta}}\partial_\vartheta Y + \widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_\varphi Y)]_s && (88)\\
&= [\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}\partial_{\vartheta\vartheta}Y - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\vartheta}\partial_{\varphi\vartheta}Y + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\partial_{\vartheta\varphi}\frac{1}{\sin\vartheta}Y - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y && (89)\\
&+ (\widehat{\boldsymbol{\varphi}}\partial_\vartheta\widehat{\boldsymbol{\vartheta}} - \frac{1}{\sin\vartheta}\widehat{\boldsymbol{\vartheta}}\partial_\varphi\widehat{\boldsymbol{\vartheta}})\partial_\vartheta Y + (\frac{1}{\sin\vartheta}\widehat{\boldsymbol{\varphi}}\partial_\vartheta\widehat{\boldsymbol{\varphi}} - \frac{1}{\sin^2\vartheta}\widehat{\boldsymbol{\vartheta}}\partial_\varphi\widehat{\boldsymbol{\varphi}})\partial_\varphi Y]_s && (90)\\
&= [\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}\partial_{\vartheta\vartheta}Y - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\vartheta}\partial_{\varphi\vartheta}Y + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_{\vartheta\varphi}Y - \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\frac{\cos\vartheta}{\sin^2\vartheta}\partial_\varphi Y - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y && (91)\\
&+ (-\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{r}} - \frac{\cos\vartheta}{\sin\vartheta}\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}})\partial_\vartheta Y + \frac{1}{\sin^2\vartheta}\widehat{\boldsymbol{\vartheta}}(\sin\vartheta\widehat{\boldsymbol{r}} + \cos\vartheta\widehat{\boldsymbol{\vartheta}})\partial_\varphi Y]_s && (92)\\
&= [\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}]_s(\partial_{\vartheta\vartheta}Y - \frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y - \frac{\cos\vartheta}{\sin\vartheta}\partial_\vartheta Y) + (\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}} - \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}})(-\frac{1}{\sin\vartheta}\partial_{\varphi\vartheta}Y + \frac{\cos\vartheta}{\sin^2\vartheta}\partial_\varphi Y) && (93)\\
&- [\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{r}}]_s\partial_\vartheta Y + [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{r}}]_s\frac{1}{\sin\vartheta}\partial_\varphi Y && (94)
\end{aligned}$$

This is semi-consistent with what's written in Eq. 57, not counting the $\sin\vartheta$ discrepancies. The last two terms can be cancelled out by adding $\underline{\underline{\mathrm{c}}}$, as written, to make it completely tangential:

$$\underline{\underline{\mathrm{d}}} = [\boldsymbol{L}(r\nabla)Y]_s + \underline{\underline{\mathrm{c}}} \tag{95}$$

Next:

$$\begin{aligned}
[(r\nabla)(r\nabla)Y]_s &= [(r\nabla)(\widehat{\boldsymbol{\vartheta}}\partial_\vartheta Y + \widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_\varphi Y)]_s && (96)\\
&= [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}\partial_{\vartheta\vartheta}Y + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\vartheta}\partial_{\varphi\vartheta}Y + \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}\partial_{\vartheta\varphi}\frac{1}{\sin\vartheta}Y + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y && (97)\\
&+ (\widehat{\boldsymbol{\vartheta}}\partial_\vartheta\widehat{\boldsymbol{\vartheta}} + \frac{1}{\sin\vartheta}\widehat{\boldsymbol{\varphi}}\partial_\varphi\widehat{\boldsymbol{\vartheta}})\partial_\vartheta Y + (\frac{1}{\sin\vartheta}\widehat{\boldsymbol{\vartheta}}\partial_\vartheta\widehat{\boldsymbol{\varphi}} + \frac{1}{\sin^2\vartheta}\widehat{\boldsymbol{\varphi}}\partial_\varphi\widehat{\boldsymbol{\varphi}})\partial_\varphi Y]_s && (98)\\
&= [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}\partial_{\vartheta\vartheta}Y + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\vartheta}\partial_{\varphi\vartheta}Y + \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_{\vartheta\varphi}Y - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}\frac{\cos\vartheta}{\sin^2\vartheta}\partial_\varphi Y + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y && (99)\\
&+ (-\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{r}} + \frac{\cos\vartheta}{\sin\vartheta}\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}})\partial_\vartheta Y - \frac{1}{\sin^2\vartheta}\widehat{\boldsymbol{\varphi}}(\sin\vartheta\widehat{\boldsymbol{r}} + \cos\vartheta\widehat{\boldsymbol{\vartheta}})\partial_\varphi Y]_s && (100)\\
&= \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}\partial_{\vartheta\vartheta}Y + [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}]_s(2\frac{1}{\sin\vartheta}\partial_{\varphi\vartheta}Y - 2\frac{\cos\vartheta}{\sin^2\vartheta}\partial_\varphi Y) + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}(\frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y + \frac{\cos\vartheta}{\sin\vartheta}\partial_\vartheta Y) && (101)\\
&- [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{r}}]_s\partial_\vartheta Y - \frac{1}{\sin\vartheta}[\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{r}}]_s\partial_\varphi Y && (102)
\end{aligned}$$

The $\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}$ term can be transformed by $\nabla^2 = \partial_{\vartheta\vartheta} + \frac{\cos\vartheta}{\sin\vartheta}\partial_\theta + \frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}$ into the expression (7a) in Zerilli. Also, $\sin\vartheta$ discrepancy remains. In our perameterization, we define:

$$\underline{\underline{\mathrm{e}}} = [(r\nabla)(r\nabla)Y]_s + \underline{\underline{\mathrm{b}}} \tag{103}$$

The last one:

$$[\boldsymbol{L}\boldsymbol{L}Y]_s = [(r\nabla)(\widehat{\boldsymbol{\varphi}}\partial_\vartheta Y - \widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\vartheta}\partial_\varphi Y)]_s \tag{104}$$

$$= [\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\partial_{\vartheta\vartheta}Y - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_{\varphi\vartheta}Y - \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}\partial_{\vartheta\varphi}\frac{1}{\sin\vartheta}Y + \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}\frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y \tag{105}$$

$$+ (\widehat{\boldsymbol{\varphi}}\partial_\vartheta\widehat{\boldsymbol{\varphi}} - \frac{1}{\sin\vartheta}\widehat{\boldsymbol{\vartheta}}\partial_\varphi\widehat{\boldsymbol{\varphi}})\partial_\vartheta Y - (\frac{1}{\sin\vartheta}\widehat{\boldsymbol{\varphi}}\partial_\vartheta\widehat{\boldsymbol{\vartheta}} - \frac{1}{\sin^2\vartheta}\widehat{\boldsymbol{\vartheta}}\partial_\varphi\widehat{\boldsymbol{\vartheta}})\partial_\varphi Y]_s \tag{106}$$

$$= [\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\partial_{\vartheta\vartheta}Y - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}\frac{1}{\sin\vartheta}\partial_{\varphi\vartheta}Y - \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\vartheta}\partial_{\vartheta\varphi}Y + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}\frac{\cos\vartheta}{\sin^2\vartheta}\partial_\varphi Y + \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}\frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y \tag{107}$$

$$+ \widehat{\boldsymbol{\vartheta}}\frac{1}{\sin\vartheta}(\sin\vartheta\widehat{\boldsymbol{r}} + \cos\vartheta\widehat{\boldsymbol{\vartheta}})\partial_\vartheta Y - (-\frac{1}{\sin\vartheta}\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{r}} - \frac{\cos\vartheta}{\sin^2\vartheta}\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}})\partial_\varphi Y]_s \tag{108}$$

$$= \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}(\frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y + \frac{\cos\vartheta}{\sin\vartheta}\partial_\theta Y) + [\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\vartheta}}]_s(-2\frac{1}{\sin\vartheta}\partial_{\vartheta\varphi}Y + 2\frac{\cos\vartheta}{\sin^2\vartheta}\partial_\varphi Y) + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\partial_{\vartheta\vartheta}Y \tag{109}$$

$$+ [\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{r}}]_s\frac{1}{\sin\vartheta}\partial_\varphi Y + [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{r}}]_s\partial_\vartheta Y \tag{110}$$

Again, we get

$$\underline{\underline{\mathrm{h}}} = [\boldsymbol{L}\boldsymbol{L}Y]_s + \underline{\underline{\mathrm{b}}} \tag{111}$$

Careful, the symmetrized off-diagonal terms in this definition have $\frac{1}{2}$ elements.

The discrepancy in $\widehat{\boldsymbol{\varphi}}$ components matches the metric. It is a consequence of the fact that we have upper and lower indices, and a metric tensor. If we want the components of the tensor to match a Cartesian tensor on the surface of the sphere, we need to follow our own expressions.

The whole set:

$$X = 2\left[\frac{1}{\sin\vartheta}\partial_{\vartheta\varphi}Y - \frac{\cos\vartheta}{\sin^2\vartheta}\partial_\varphi Y\right] \tag{112}$$

$$W = \partial_{\vartheta\vartheta}Y - \frac{1}{\sin^2\vartheta}\partial_{\varphi\varphi}Y - \frac{\cos\varphi}{\sin\varphi}\partial_\vartheta Y = 2\partial_{\vartheta\vartheta}Y + l(l+1)Y \tag{113}$$

$$Z = -\partial_{\vartheta\vartheta}Y - l(l+1)Y \tag{114}$$

$$E = \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}\partial_{\vartheta\vartheta}Y + [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}]_s X + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}Z \tag{115}$$

$$H = \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}}\partial_{\vartheta\vartheta}Y - [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}]_s X + \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}}Z \tag{116}$$

$$t^{(0)} = \widehat{\boldsymbol{r}}\widehat{\boldsymbol{r}}Y \tag{117}$$

$$t^{(1)} = \sqrt{\frac{2}{l(l+1)}}\left\{[\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\vartheta}}]_s\partial_\vartheta Y + \frac{1}{\sin\vartheta}[\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}]_s\partial_\varphi Y\right\} = C[\widehat{\boldsymbol{r}}\boldsymbol{\Psi}]_s \tag{118}$$

$$t^{(2)} = \sqrt{\frac{2}{l(l+1)}}\left\{[\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\varphi}}]_s\partial_\vartheta Y - \frac{1}{\sin\vartheta}[\widehat{\boldsymbol{r}}\widehat{\boldsymbol{\vartheta}}]_s\partial_\varphi Y\right\} = C[\widehat{\boldsymbol{r}}\boldsymbol{\Phi}]_s \tag{119}$$

$$t^{(3)} = \sqrt{\frac{2}{(l-1)l(l+1)(l+2)}}\left\{\tfrac{1}{2}(\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}} - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}})X + [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}]_s W\right\} \tag{120}$$

$$t^{(4)} = \frac{1}{\sqrt{2}l(l+1)}\{E+H\} = \frac{1}{\sqrt{2}}\left\{-(\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}} + \widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}})Y\right\} \tag{121}$$

$$t^{(5)} = \sqrt{\frac{1}{2(l-1)l(l+1)(l+2)}}\{E-H\} = \sqrt{\frac{2}{(l-1)l(l+1)(l+2)}}\left\{-\tfrac{1}{2}(\widehat{\boldsymbol{\varphi}}\widehat{\boldsymbol{\varphi}} - \widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\vartheta}})W + [\widehat{\boldsymbol{\vartheta}}\widehat{\boldsymbol{\varphi}}]_s X\right\} \tag{122}$$

These are all orthonormalized in the sense

$$\oint t^i : t^j \mathrm{d}\Omega = \delta_{ij}. \tag{123}$$

in addition, they are pointwise orthogonal.

This simplification exhibits much more symmetry than anything before it. The tensors $t^{(0)}$ and $t^{(4)}$ are diagonal and have a nonzero trace (the radial and tangential part). For a traceless base, we need

$$t^{(D)} = \sqrt{\frac{2}{3}}t^{(0)} + \sqrt{\frac{1}{3}}t^{(4)}. \tag{124}$$

$t^{(3)}$ and $t^{(5)}$ show explicit symmetry in the tangent plane. It represents the local curvature of Y and its orthogonal conjugate. The radial parts are directly related to vector harmonics.

To research: what is $\nabla^2 \underline{\underline{t}}^i$ and is it nicely separable?

IV. RESPONSE OF HARMONICS TO DIFFERENTIAL OPERATORS

We will investigate action of four operators on the harmonics:

$$\boldsymbol{L}\square = \boldsymbol{r} \times \nabla\square = ? \tag{125}$$
$$(r\nabla)\square = -(\widehat{\boldsymbol{r}} \times \boldsymbol{L})\square = ? \tag{126}$$
$$r\nabla \times \square = ? \tag{127}$$
$$r^2\nabla^2\square = (\partial_{rr} + 2\partial_r + \boldsymbol{L}^2)\square = ? \tag{128}$$

With this, we hope solve certain differential equations.

A. Action on scalar harmonics

Scalar harmonics are simple, because they just generate the vector harmonics:

$$(r\nabla)Y_{lm} = \sqrt{l(l+1)}\boldsymbol{\Psi}_{lm} \tag{129}$$
$$\boldsymbol{r} \times \nabla Y_{lm} = \sqrt{l(l+1)}\boldsymbol{\Phi}_{lm} \tag{130}$$
$$r^2\nabla^2 Y_{lm} = -l(l+1)Y_{lm} \tag{131}$$

B. Action on vector harmonics

Divergence:

$$r\nabla \cdot \boldsymbol{Y}_{lm} = r\widehat{\boldsymbol{r}} \cdot \nabla Y_{lm} + rY_{lm}\nabla \cdot \widehat{\boldsymbol{r}} = 2Y_{lm} \tag{132}$$
$$r\nabla \cdot \boldsymbol{\Psi}_{lm} = \frac{1}{\sqrt{l(l+1)}}r\nabla \cdot (r\nabla)Y_{lm} = \frac{1}{\sqrt{l(l+1)}}\nabla^2 Y_{lm} = -\sqrt{l(l+1)}Y_{lm} \tag{133}$$
$$r\nabla \cdot \boldsymbol{\Phi}_{lm} = \frac{1}{\sqrt{l(l+1)}}r\nabla \cdot (r\widehat{\boldsymbol{r}} \times \nabla)Y_{lm} = 0 \tag{134}$$

Curl:

$$r\nabla \times \boldsymbol{Y}_{lm} = r(\nabla \times \widehat{\boldsymbol{r}})Y_{lm} + r(\nabla Y_{lm}) \times \widehat{\boldsymbol{r}} = -\sqrt{l(l+1)}\boldsymbol{\Phi}_{lm} \tag{135}$$
$$r\nabla \times \boldsymbol{\Psi}_{lm} = \frac{1}{\sqrt{l(l+1)}}r\nabla \times (r\nabla Y_{lm}) = \frac{1}{\sqrt{l(l+1)}}r(\nabla r) \times (\nabla Y_{lm}) = \boldsymbol{\Phi}_{lm} \tag{136}$$
$$r\nabla \times \boldsymbol{\Phi}_{lm} = \frac{1}{\sqrt{l(l+1)}}r\nabla \times \boldsymbol{r} \times \nabla Y_{lm} = \frac{1}{\sqrt{l(l+1)}}\left[r\boldsymbol{r}\nabla^2 Y_{lm} - r(\boldsymbol{r} \cdot \nabla)\nabla Y_{lm} + r(\nabla Y_{lm}\nabla)\boldsymbol{r} - r(\nabla \cdot \boldsymbol{r})\nabla Y_{lm}\right] = \tag{137}$$
$$= -\sqrt{l(l+1)}\boldsymbol{Y}_{lm} - \boldsymbol{\Psi}_{lm} \tag{138}$$

$$r\nabla \times r^n\boldsymbol{Y}_{lm} = -\sqrt{l(l+1)}r^n\boldsymbol{\Phi}_{lm} \tag{139}$$
$$r\nabla \times r^n\boldsymbol{\Psi}_{lm} = (n+1)r^n\boldsymbol{\Phi}_{lm} \tag{140}$$
$$r\nabla \times r^n\boldsymbol{\Phi}_{lm} = r^n(-\sqrt{l(l+1)}\boldsymbol{Y}_{lm} - (n+1)\boldsymbol{\Psi}_{lm}) \tag{141}$$

Elle:

$$\boldsymbol{L}\cdot\boldsymbol{Y}_{lm} = \boldsymbol{r}\cdot(\nabla\times\boldsymbol{Y}_{lm}) = 0 \tag{142}$$
$$\boldsymbol{L}\cdot\boldsymbol{\Psi}_{lm} = \boldsymbol{r}\cdot(\nabla\times\boldsymbol{\Psi}_{lm}) = 0 \tag{143}$$
$$\boldsymbol{L}\cdot\boldsymbol{\Phi}_{lm} = \boldsymbol{r}\cdot(\nabla\times\boldsymbol{\Phi}_{lm}) = -\sqrt{l(l+1)}Y_{lm} \tag{144}$$

Laplace:

$$r^2\nabla^2\boldsymbol{Y}_{lm} = -(l^2+l+2)\boldsymbol{Y}_{lm} + 2\sqrt{l(l+1)}\boldsymbol{\Psi}_{lm} \tag{145}$$
$$r^2\nabla^2\boldsymbol{\Psi}_{lm} = 2\sqrt{l(l+1)}\boldsymbol{Y}_{lm} - l(l+1)\boldsymbol{\Psi}_{lm} \tag{146}$$
$$r^2\nabla^2\boldsymbol{\Phi}_{lm} = l(l+1)\boldsymbol{\Phi}_{lm} \tag{147}$$

Here, we notice the splitting (VSH are not eigenfuctions of the Laplacian/Casimir):

$$\boldsymbol{Z}^{(l+1)}_{lm} = -\sqrt{l(l+1)}\boldsymbol{Y}_{lm} + l\boldsymbol{\Psi}_{lm} \tag{148}$$
$$\boldsymbol{Z}^{(l)}_{lm} = l\boldsymbol{Y}_{lm} + \sqrt{l(l+1)}\boldsymbol{\Psi}_{lm} \tag{149}$$
$$r^2\nabla^2\boldsymbol{Z}^{(l+1)}_{lm} = -(l+1)(l+2)\boldsymbol{Z}^{(l+1)}_{lm} \tag{150}$$
$$r^2\nabla^2\boldsymbol{Z}^{(l)}_{lm} = -l(l+1)\boldsymbol{Z}^{(l)}_{lm} \tag{151}$$

C. Action on tensor harmonics

Basis for this study is the traceless set: t^r, t^E, t^B, $t^\times$, t^+. The Laplacian braket, in matrix form:

$$\langle t^i|r^2\nabla^2|t^j\rangle = \begin{bmatrix} -l(l+1)-6 & \sqrt{12l(l+1)} & 0 & 0 & 0 \\ \sqrt{12l(l+1)} & -l(l+1)-4 & 0 & 0 & 2\sqrt{l(l+1)-2} \\ 0 & 0 & -l(l+1)-4 & 2\sqrt{l(l+1)-2} & 0 \\ 0 & 0 & 2\sqrt{l(l+1)-2} & 2-l(l+1) & 0 \\ 0 & 2\sqrt{l(l+1)-2} & 0 & 0 & 2-l(l+1) \end{bmatrix} \tag{152}$$

We get a mix of l, $l+1$ and $l+2$ eigentensors. None of the states are pure.

Diagonalizing this matrix gives us a mapping from our set of tensor harmonics to a set of proper eigenvalues and eigentensors of the Laplacian:

$$\lambda = \{-(\ell+2)(\ell+3), -(\ell+1)(\ell+2), -\ell(\ell+1), -(\ell-1)\ell, -(\ell-2)(\ell-1)\} \tag{153}$$

$$P = \begin{bmatrix} \frac{\sqrt{3}\sqrt{\ell(\ell+1)}\sqrt{-2+\ell+\ell^2}}{\ell(\ell-1)} & 0 & -\frac{\sqrt{\ell(\ell+1)}}{\sqrt{3}\sqrt{-2+\ell+\ell^2}} & 0 & \frac{\sqrt{3}\sqrt{\ell(\ell+1)}\sqrt{-2+\ell+\ell^2}}{(\ell+1)(\ell+2)} \\ -\frac{2\sqrt{-2+\ell+\ell^2}}{\ell-1} & 0 & -\frac{1}{\sqrt{-2+\ell+\ell^2}} & 0 & \frac{2\sqrt{-2+\ell+\ell^2}}{\ell+2} \\ 0 & -\frac{\sqrt{-2+\ell+\ell^2}}{\ell-1} & 0 & \frac{\sqrt{-2+\ell+\ell^2}}{\ell+2} & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} \tag{154}$$

The columns correspond to linear combinations that result in corresponding eigenvalues.

V. A CASE STUDY: NEMATIC DROPLETS

In nematic droplets [7], we are looking for solutions to

$$\boldsymbol{m}(\nabla\times\boldsymbol{m}) = -q_0|\boldsymbol{m}|^2 \tag{155}$$

or rather

$$\nabla\times\boldsymbol{m} = -q_0\boldsymbol{m}. \tag{156}$$

This was solved by a harmonic function, decorated by a higher order curl term:

$$\boldsymbol{m} = \nabla\phi + \boldsymbol{m}_c \tag{157}$$

where

$$\phi = \sum a_{lm} r^l Y_{lm}, \quad \boldsymbol{m}_c = q_0 \sum \frac{a_{lm}}{l+1} r^l \boldsymbol{r} \times \nabla Y_{lm}. \tag{158}$$

This can be rephrased in terms of vector harmonics (let's ignore the trivial sum over coefficients now):

$$\nabla\phi = l r^{l-1}\widehat{\boldsymbol{r}} Y_{lm} + r^{l-1}(r\nabla)Y_{lm} = r^{l-1}(l\boldsymbol{Y}_{lm} + \sqrt{l(l+1)}\boldsymbol{\Psi}_{lm}) \tag{159}$$

$$\boldsymbol{m}_c = q_0 r^l \frac{1}{l+1}\boldsymbol{L}Y_{lm} = q_0 r^l \sqrt{\frac{l}{l+1}}\boldsymbol{\Phi}_{lm} \tag{160}$$

This fits, but this is actually a sequence of increasing order, where curl of next one gives the previous one. Next one to solve is:

$$\nabla \times \boldsymbol{m}^{(2)} = -q_0 \boldsymbol{m}^{(1)} = -q_0^2 r^l \sqrt{\frac{l}{l+1}}\boldsymbol{\Phi}_{lm} \tag{161}$$

This equation has an ambiguous solution, as both $\boldsymbol{\Psi}$ and $\boldsymbol{Y}$ yield $\boldsymbol{\Phi}$ upon application of curl. However, in order to continue the iteration, we must pick a combination that is a valid result of a curl operator itself. Therefore, according to Eq. 141, we look for the solution in the form

$$\boldsymbol{m}^{(2)} = A r^{l+1}(\sqrt{l(l+1)}\boldsymbol{Y}_{lm} + (l+3)\boldsymbol{\Psi}_{lm}) \tag{162}$$

$$r\nabla \times \boldsymbol{m}^{(2)} = A r^{l+1}(-l(l+1) + (l+2)(l+3))\boldsymbol{\Phi}_{lm} = A r^{l+1}(4l+6)\boldsymbol{\Phi}_{lm} \tag{163}$$

By comparison, we get $A = -q_0^2\sqrt{l/(l+1)}/(4l+6)$. Because we set up the term correctly, next iteration is again pure curl harmonic:

$$\boldsymbol{m}^{(3)} = q_0 A r^{l+2}\boldsymbol{\Phi}_{lm} \tag{164}$$

So we get the following sequence:

$$\boldsymbol{m}^{(0)} = r^{l-1}(l\boldsymbol{Y}_{lm} + \sqrt{l(l+1)}\boldsymbol{\Psi}_{lm}) \tag{165}$$

$$\boldsymbol{m}^{(1)} = q_0 r^l \sqrt{\frac{l}{l+1}}\boldsymbol{\Phi}_{lm} \tag{166}$$

$$\boldsymbol{m}^{(2)} = -q_0^2 r^{l+1}(\frac{l}{4l+6}\boldsymbol{Y}_{lm} + \frac{\sqrt{l/(l+1)}(l+3)}{4l+6}\boldsymbol{\Psi}_{lm}) \tag{167}$$

$$\boldsymbol{m}^{(3)} = -q_0^3 r^{l+2}\frac{\sqrt{l/(l+1)}}{4l+6}\boldsymbol{\Phi}_{lm} \tag{168}$$

Even better, we can write it down recursively. Odd terms are curly. Even terms are of the form:

$$\boldsymbol{m}^{(2k)} = q_0^{2k} r^{l-1+2k} a_k(\sqrt{l(l+1)}\boldsymbol{Y}_{lm} + (l+2k+1)\boldsymbol{\Psi}_{lm}), \tag{169}$$

and odd terms are

$$\boldsymbol{m}^{(2k+1)} = q_0^{2k+1} r^{l+2k} a_k \boldsymbol{\Phi}_{lm}. \tag{170}$$

We get a recursion for the pre-factors, and the initial condition:

$$a_{k+1} = -a_k \frac{1}{2(k+1)(3+2l+2k)}, \quad a_0 = \sqrt{\frac{l}{l+1}} \tag{171}$$

$$a_k = (-1)^k a_0 \frac{(1+2l)!!}{2^k k!(1+2l+2k)!!} \tag{172}$$

Summation:

$$\boldsymbol{m} = r^{l-1}(1+2l)!!\sqrt{\frac{l}{l+1}}\sum_{k=0}^{\infty}\frac{1}{k!(1+2l+2k)!!}\left(-\tfrac{1}{2}(q_0 r)^2\right)^k\left[\sqrt{l(l+1)}\boldsymbol{Y}_{lm} + q_0 r\boldsymbol{\Phi}_{lm} + (l+2k+1)\boldsymbol{\Psi}_{lm}\right] = \tag{173}$$

$$= r^{l-1}\sqrt{\frac{l}{l+1}}\Big[{}_0F_1\left(l+\tfrac{3}{2}, -\tfrac{(rq_0)^2}{4}\right)\left(\sqrt{l(l+1)}\boldsymbol{Y}_{lm} + q_0 r\boldsymbol{\Phi}_{lm} + (l+1)\boldsymbol{\Psi}_{lm}\right) \tag{174}$$

$$- \sqrt{\pi/2}(q_0 r)^{1/2-l}(2l+1)!!J_{l+3/2}(q_0 r)\boldsymbol{\Psi}_{lm}\Big] \tag{175}$$

$$= r^{l-1}\sqrt{\frac{l}{l+1}}\Big[J_{l+1/2}(q_0 r)\Gamma(l+3/2)(q_0 r/2)^{-l-1/2}\left(\sqrt{l(l+1)}\boldsymbol{Y}_{lm} + q_0 r\boldsymbol{\Phi}_{lm} + (l+1)\boldsymbol{\Psi}_{lm}\right) \tag{176}$$

$$- (2l+1)!!(q_0 r)^{1-l} j_{l+1}(q_0 r)\boldsymbol{\Psi}_{lm}\Big] \tag{177}$$

$$= (2l+1)!!r^{l-1}\Big[\frac{j_l(q_0 r)}{(q_0 r)^l}\left(l\boldsymbol{Y}_{lm} + q_0 r\sqrt{\frac{l}{l+1}}\boldsymbol{\Phi}_{lm} + \sqrt{l(l+1)}\boldsymbol{\Psi}_{lm}\right) - \sqrt{\frac{l}{l+1}}\frac{j_{l+1}(q_0 r)}{(q_0 r)^{l+1}}\boldsymbol{\Psi}_{lm}\Big] \tag{178}$$

where j is the spherical Bessel function of the first kind. This is a closed form solution with constant chirality throughout the entire sample, not just around the origin.

VI. FLUCTUATIONS

Consider a set of particles on a sphere and its expansion to ℓm coefficients:

$$\rho_{\ell m} = \sum_k Y^*_{\ell m}(\Omega_k). \tag{179}$$

If the particles move, each in its own tangential direction $\boldsymbol{\mu}_k$, we can calculate the perturbation to these coefficients:

$$\rho_{\ell m} = \sum_k Y^*_{\ell m}(\Omega_k + \boldsymbol{\mu}_k) \approx \sum_k Y^*_{\ell m}(\Omega_k) + \boldsymbol{\mu}_k \cdot \nabla Y^*_{\ell m}(\Omega_k) = \sum_k Y^*_{\ell m}(\Omega_k) + \sqrt{l(l+1)}\frac{\boldsymbol{\mu}_k}{r}\cdot\boldsymbol{\Psi}^*_{\ell m}(\Omega_k). \tag{180}$$

Notice a few peculiarities. First, only the gradient (electric-type) vector harmonics matter. The curl (magnetic-type) vector harmonics represent modes of motion that preserve the coefficients ("Goldstone modes")?

In fluctuations $\boldsymbol{\mu}$ has a zero mean. We need average squares of components over time, that contribute to Q_ℓ. We obtain:

$$\langle\rho^2_{\ell m}\rangle = |\rho^0_{\ell m}|^2 + l(l+1)r^{-2}\left|\sum_k \boldsymbol{\mu}_k\boldsymbol{\Psi}^*_{\ell m}(\Omega_k)\right|^2 \tag{181}$$

This holds true for each of the eigenmodes which have different amplitude vectors $\boldsymbol{\mu}_{pk}$, which are uncorrelated, so they can be added in squares.

The $\boldsymbol{\mu}_{pk}$ is a p-th eigenvector of the Hessian H. Equipartition says that we should have

$$\frac{1}{2}kT = \frac{1}{2}\boldsymbol{\mu}_p H\boldsymbol{\mu}_p = \frac{1}{2}E_p|\boldsymbol{\mu}_p|^2 \tag{182}$$

with E_p being the corresponding eigenvalue. This sets the square magnitudes of the eigenmodes. Finally, expressed with a normalized eigenvector:

$$\langle\rho^2_{\ell m}\rangle = |\rho^0_{\ell m}|^2 + kTl(l+1)r^{-2}\sum_p\frac{1}{E_p}\left|\sum_k \tilde{\boldsymbol{\mu}}_{pk}\boldsymbol{\Psi}^*_{\ell m}(\Omega_k)\right|^2 \tag{183}$$

Problem: this doesn't ever decrease the structure factor, only adds to it, due to cross-terms being zero by assumption of zero-mean incoherent modes. This is somewhat reasonable – fluctuations around the equilibrium just spread out

the particles around their positions, but do not describe breakdown of ordered structure. That is a non-perturbative effect. However, this calculation does not give the same convolutional result that we get by just spreading out the particles Gaussianily.

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[1] F. Mignard and S. Klioner, Astronomy & Astrophysics **547**, A59 (2012).
[2] A. Lošdorfer Božič and S. Čopar, Phys. Rev. E **99**, 032601 (2019), arXiv:1812.11763 [cond-mat.soft].
[3] R. G. Barrera, G. Estevez, and J. Giraldo, European Journal of Physics **6**, 287 (1985).
[4] B. Carrascal, G. Estevez, P. Lee, and V. Lorenzo, European Journal of Physics **12**, 184 (1991).
[5] K. S. Thorne, Reviews of Modern Physics **52**, 299 (1980).
[6] F. J. Zerilli, Journal of Mathematical Physics **11**, 2203 (1970).
[7] J. Pollard, G. Posnjak, S. Čopar, I. Muševič, and G. P. Alexander, Phys. Rev. X **9**, 1442 (2019).